C4.4 Why are nanoparticles so useful?		
Teaching and learning narrative	**Assessable learning outcomes** *Learners will be required to:*	***Linked learning opportunities***
Nanoparticles have a similar scale to individual molecules. Their extremely small size means they can penetrate into biological tissues and can be incorporated into other materials to modify their properties. Nanoparticles have a very high surface area to volume ratio. This makes them excellent catalysts. Fullerenes form nanotubes and balls. The ball structure enables them to carry small molecules, for example carrying drugs into the body. The small size of fullerene nanotubes enables them to be used as molecular sieves and to be incorporated into other materials (for example to increase strength of sports equipment). Graphene sheets have specialised uses because they are only a single atom thick but are very strong with high electrical and thermal conductivity. Developing technologies based on fullerenes and graphene required leaps of imagination from creative thinkers (IaS3).	1. compare 'nano' dimensions to typical dimensions of atoms and molecules 2. describe the surface area to volume relationship for different-sized particles and describe how this affects properties 3. describe how the properties of nanoparticulate materials are related to their uses including properties which arise from their size, surface area and arrangement of atoms in tubes or rings 4. explain the properties of fullerenes and graphene in terms of their structures 5. explain the possible risks associated with some nanoparticulate materials including: a) possible effects on health due to their size and surface area b) reasons that there is more data about uses of nanoparticles than about possible health effects c) the relative risks and benefits of using nanoparticles for different purposes	**Ideas about Science:** • Discuss the potential benefits and risks of developments in nanotechnology (IaS4) • Development of nanoparticles and graphene relied on imaginative thinking. (IaS3)

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There are concerns about the safety of some nanoparticles because not much is known about their effects on the human body. Judgements about a particular use for nanoparticles depend on balancing the perceived benefit and risk (IaS4).	6. estimate size and scale of atoms and nanoparticles including the ideas that: a) nanotechnology is the use and control of structures that are very small (1 to 100 nanometres in size) b) data expressed in nanometres is used to compare the sizes of nanoparticles, atoms and molecules M1d	
	7. interpret, order and calculate with numbers written in standard form when dealing with nanoparticles M1b	
	8. use ratios when considering relative sizes and surface area to volume comparisons M1c	
	9. calculate surface areas and volumes of cubes M5c	